

Design and Implementation of Distributed Applications

dida-tkvs

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Abstract

dida-tkvs is a replicated key-value store composed of multiple servers, transactional clients, and a configuration console. High-performance consistency across replicas is achieved using the Multi-Paxos consensus algorithm. In this paper, we focus on the most critical aspects of our implementation.

1. Introduction

In distributed systems, ensuring consistency and coordination across multiple nodes is a challenging task, especially in environments where failures can occur. Paxos is a consensus algorithm that addresses these issues by allowing a set of nodes to agree on a single value, even in the presence of failures. In this project, we implemented a transactional key-value store replicated using Paxos to ensure that all servers maintain a consistent view of the data.

The system we developed operates in phases, starting from basic client-server communication and evolving into a fully reconfigurable Paxos-based system. This paper highlights the key components of the system, focusing on the main loop that drives the consensus process and handles new client requests. The logic behind selecting the next round number for the leader is also discussed, showcasing how our system ensures liveness and consistency.

2. System Overview

The key-value store consists of multiple servers, each holding a replicated state of the store. Clients initiate transactions, which involve reading and updating key-value pairs. Servers coordinate through Paxos to assign unique timestamps to updates, ensuring that conflicting operations are managed correctly. The system can dynamically reconfigure itself, allowing different sets of servers to take on

the roles of proposers and acceptors while ensuring that all nodes act as learners.

2.1. Main Loop for Transaction Proposals and Execution

The main loop is central to the system's operation, where the leader proposes transactions, and the learners execute them. The flow is as follows:

- Transaction Proposal by Leader: To be completed.
- Transaction Execution by Learners: To be completed.

2.2. Handling New Requests

When a new request is received from a client, the system must ensure that the transaction is processed in a consistent and timely manner. This involves coordinating between the leader and the learners to ensure the request is either committed or aborted based on the current state of the key-value store. To be completed.

2.3. Computing the Next Round Number (rnd) for the Leader

The leader plays a crucial role in managing the order of transactions. One of its key responsibilities is determining the next round number (rnd) to use in Paxos. This ensures that no conflicting proposals are made, and that progress is always possible. To be completed.

3. Conclusion

The implementation of our transactional key-value store using Paxos demonstrates the effectiveness of the algorithm in maintaining consistency across distributed servers. The main loop and its handling of transaction proposals and executions are vital to the system's overall functionality. While

we have discussed key aspects of the system, including how the leader computes the next round number, further optimization could be explored, especially in handling client requests efficiently. Overall, our implementation provides a robust foundation for distributed consensus in transactional systems.

Figure 1. Example of caption.

References

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